NextHikes IT Solutions

Bike Sharing Demand Analysis Project Using Excel

Preprocess Summary Report

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Project 1 - Data Set 1 Preprocess Report

Preprocessing Summary Report	
Step	Detail
Original Shape	610 rows × 10 columns
After Preprocessing	610 rows × 11 columns
Duplicates Found	0 (None removed)
Missing Values	0 in all columns

Transformations Applied	
season	1→Spring, 2→Summer, 3→Fall, 4→Winter
yr	0→2011, 1→2012
weathersit	1→Clear, 2→Mist/Cloudy, 3→Light Rain/Snow, 4→Heavy Rain/Snow
holiday	True/False → Yes/No
weekday	0→Sunday … 6→Saturday
mnth	1→Jan … 12→Dec
temp_celsius (new col)	Temp_Normalized × 41 → Actual °C
dteday	Datetime → Date only (YYYY-MM-DD)

Column Renames	
instant → Record_ID	dteday → Date
season → Season	yr → Year
mnth → Month	hr → Hour
holiday → Is_Holiday	weekday → Weekday
weathersit → Weather_Situation	temp → Temp_Normalized

Project 2 - Data Set 2 Preprocess Report

OVERVIEW	
Original Shape	610 rows × 15 columns
Final Shape	610 rows × 10 columns
STEP 1 — Junk Column Removal	
Columns Removed	8 unwanted columns dropped
'Unnamed: 0'	Redundant row index copy
'Unnamed: 8/12'	Completely empty columns (610/610 NaN)
'Mean','Mode','Median'	Completely empty columns (610/610 NaN)
'0.2879' + 'Unnamed: 14'	Stats sidebar accidentally embedded in sheet

STEP 2 — Duplicate Removal	
Duplicates Found & Dropped	0
STEP 3 — Missing Value Treatment	
'atemp' column	11 nulls filled with median = 0.197
Strategy	Median imputation
STEP 4 — Numeric Inconsistency Fix	
'hum' = 1.0 (100% humidity)	1 row(s) capped to 0.99 (sensor anomaly)

STEP 5 — Denormalization (New Columns)	
'atemp_celsius'	Feels_Like_Temp_Norm × 50 → °C
'humidity_pct'	Humidity_Norm × 100 → %
'windspeed_kmh'	Windspeed_Norm × 67 → km/h
STEP 6 — Column Renames	
instant → Record_ID	atemp → Feels_Like_Temp_Norm
hum → Humidity_Norm	windspeed → Windspeed_Norm
casual → Casual_Users	registered → Registered_Users
cnt → Total_Users	

Project 3 - Data Set 3 Preprocess Report

DATASET OVERVIEW	
Original Shape	390 rows × 16 columns
Final Shape	390 rows × 17 columns
Date Range	2011-01-28 to 2011-02-14 (18 days)
Record_ID Range	611 – 1000 (continuation of datasets 1 & 2)
STEP 1 — Duplicate Removal	
Duplicates Found & Dropped	0

STEP 2 — Missing Values	
Missing Values Found	0 (dataset structurally complete)
STEP 3 — Zero-Variance Column Drop	
'season' dropped	All 390 rows = 1 (Spring only) — no analytical value
'yr' dropped	All 390 rows = 0 (year 2011 only) — no analytical value
'holiday' dropped	All 390 rows = False — no variation in this slice

STEP 4 — Humidity Anomaly Fix	
hum = 1.0 rows found	15 rows across 4 dates (sensor saturation)
Fix Strategy	Each anomalous value → daily median of valid (non-1.0) hum for that date
Anomaly Details	See 'Anomaly Log' sheet for full record-level breakdown
STEP 5 — Valid Edge Cases (Retained)	
windspeed = 0	75 rows retained — calm-wind hours are valid observations
Missing hours per day	Partial-day recordings retained as-is (reflect actual sensor gaps)

STEP 6 — Categorical Decoding	
mnth	1–12 → Jan – Dec
weathersit	1→Clear, 2→Mist/Cloudy, 3→Light Rain/Snow, 4→Heavy Rain/Snow
weekday	0→Sunday … 6→Saturday
STEP 7 — Denormalization (4 New Columns)	
temp_celsius	Temp_Norm × 41 → Actual temperature °C
atemp_celsius	FeelsLike_Norm × 50 → Feels-like temperature °C
humidity_pct	Humidity_Norm × 100 → Relative humidity %
windspeed_kmh	Windspeed_Norm × 67 → Wind speed km/h

Normalized data (0–1) is multiplied back by its **real-world maximum value** to restore original units — humidity ×100 (max 100%), windspeed ×67 (max 67 km/h), temp ×41 (max 41°C), feels-like ×50 (max 50°C).

STEP 8 — Column Renames & Reorder

instant → Record_ID	dteday → Date
mnth → Month	hr → Hour
weekday → Weekday	weathersit → Weather_Situation
temp → Temp_Norm	atemp → FeelsLike_Norm
hum → Humidity_Norm	windspeed → Windspeed_Norm
casual → Casual_Users	registered → Registered_Users
cnt → Total_Users	

COLUMN GLOSSARY	
Record_ID	Unique hourly record (611–1000, continuation of datasets 1 & 2)
Date	Observation date (YYYY-MM-DD)
Month / Hour / Weekday	Temporal labels
Weather_Situation	Clear / Mist/Cloudy / Light Rain/Snow / Heavy Rain/Snow
Temp_Norm / FeelsLike_Norm	Normalized values (0–1 scale)
temp_celsius / atemp_celsius	Human-readable temperature in °C
humidity_pct / windspeed_kmh	Humidity % and Wind speed km/h
Casual / Registered / Total_Users	Bike rental user counts per hour

Data Set A (Merge Project 1+2)

MERGE DETAILS	
Merge Key	Record_ID (inner join)
Dataset 1 Shape	610 rows × 11 columns
Dataset 2 Shape	610 rows × 10 columns (2 header rows fixed)
After Merge	610 rows × 20 columns
Final Shape	610 rows × 16 columns
STEP 1 — Duplicate Removal	
Duplicates Found	0

STEP 2 — Missing Value Treatment	
Missing Values Found	0 (none after merge)
Strategy	Median imputation for numeric columns
STEP 3 — Redundant Column Removal	
'Temp_Normalized'	Kept Temp_Celsius (human-readable °C)
'Feels_Like_Temp_Norm'	Kept atemp_celsius → renamed FeelsLike_Temp_C
'Humidity_Norm'	Kept Humidity_Pct (% form is clearer)
'Windspeed_Norm'	Kept windspeed after recomputing km/h
'windspeed_kmh' (old)	Dropped — had integer truncation error; recomputed as float

STEP 4 — Numeric Fix	
Windspeed_kmh	Recomputed: Windspeed_Norm × 67 → float (was int-truncated)
Date column	Datetime stripped to Date only (YYYY-MM-DD)
STEP 5 — Column Reordering & Rename	
Final Column Order	ID → Date/Time → Day Info → Weather → Climate → Users
atemp_celsius	→ FeelsLike_Temp_C
humidity_pct	→ Humidity_Pct

FINAL COLUMN GLOSSARY	
Record_ID	Unique hourly record identifier
Date	Date of observation (YYYY-MM-DD)
Year / Month / Season	Temporal labels (2011/2012, Jan–Dec, Spring–Winter)
Hour	Hour of day (0–23)
Weekday	Day name (Sunday–Saturday)
Is_Holiday	Yes / No
Weather_Situation	Clear / Mist/Cloudy / Light Rain/Snow / Heavy Rain/Snow
Temp_Celsius	Actual temperature in °C
FeelsLike_Temp_C	Feels-like temperature in °C
Humidity_Pct	Relative humidity in %
Windspeed_kmh	Wind speed in km/h
Casual_Users	Non-registered bike rental users
Registered_Users	Registered bike rental users
Total_Users	Total rentals (Casual + Registered)

Peak_Hour (Peak/Off-Peak)	
7 AM – 9 AM → Morning office/school rush	
5 PM – 7 PM → Evening office return	

Data Set B (Merge Data Set A + Project 3)

SOURCE FILES	
File 1	Data_Set_A.xlsx → 610 rows
File 2	Project_3.xlsx → 390 rows
Merge Strategy	Vertical concat (no overlapping IDs)
SHAPE	
After Merge	1,000 rows × 16 columns
After Preprocessing	1,000 rows × 16 columns

MISSING VALUES	
Missing Before	0
Strategy	Median imputation (numeric), "No"/"Clear" (categorical)
Missing After	0
DUPLICATES	
Duplicates Found	0
Duplicates Dropped	0

STRING / CATEGORICAL FIXES	
Columns Cleaned	Month, Season, Weekday, Is_Holiday, Weather_Situation
Operations	strip whitespace, title-case, variant mapping
Weather Variants Unified	4 categories standardised
NUMERIC CONSISTENCY FIXES	
Humidity_Pct	Clipped to valid range 0–100
User Counts	Negative values clipped to 0
Total_Users	Recomputed as Casual + Registered

NEW COLUMNS ADDED (from P3)	
Year	Extracted from Date
Season	Derived from Month (Spring/Summer/Autumn/Winter)
Is_Holiday	Defaulted to "No" for P3 subset

Thank You